

The Connectivity Railroader

Technology is expensive stuff, therefore out of reach of the common man or villagers in rural areas. Not so! Ashok Jhunjunwala and his team prove that being digital doesn't have to come at a heavy price

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Does the idea of broadband over railway cables sound very far fetched to you? How about a wireless telephone system that promises to take cost-effective, high-quality voice and data connectivity to both urban and rural areas?

These technologies aren't just confined to the insides of a lab, they are seeing the light of day thanks to Ashok Jhunjunwala and his team of talented researchers at the Telecommunications and Computer Networks group (TeNeT) at IIT, Madras. TeNeT works in areas such as Networking, Digital Systems Architecture, Fibre Optics and communication software.

Jhunjunwala, Head, Department of Electrical Engineering, is the driving force behind the

around with technology and learning a lot."

Along with his former students, he floated several R&D companies like Midas Technologies, Banyan Networks and n-Logue that work in partnership with TeNeT to develop and commercially manufacture products like corDECT Wireless in Local Loop system, Railway Signalling Network, Fibre Access Network, Remote Access Switches, Ethernet Switches, IP Switches, etc.

In the news lately is the corDECT WiLL technology, developed by TeNeT in collaboration with Midas Technologies that costs just a third of the CDMA WiLL technology. This revolutionary new technology provides voice communication using 32 Kbps ADPCM, and Internet connectivity at 35/70 Kbps. (To read more about corDECT and other technologies developed by TeNeT, head over to www.tenet.res.in). corDECT is being commercially manufactured in India under license by companies like Crompton Greaves, HFCL, etc. corDECT has been installed in India and also countries like Iran, Argentina and Brazil.

Who would ever think that the railway cables which are used to power trains could be used to power the Net? Jhunjunwala and his team along with Banyan Networks did. They hit upon an idea to use the spare quad copper cables connecting railway stations to transfer voice and data connectivity using HDSL technology. A pilot project has been initiated covering 40 km of railway track linking Vijayawada and Guntur on the South-Central Railway. Cybercafe kiosks have been built along the stretch and corDECT base stations have been deployed at Guntur and at Mangalagiri providing Internet connectivity to subscribers in these areas.

Jhunjunwala's achievements have won him several awards including the prestigious Dr Vikram Sarabhai Research Award in 1997, and more recently the Millennium Medal at Indian Science Congress in 2000. In addition, he is a Fellow of the Indian National Academy of Engineering, Indian National Science Academy and National Academy of Science and a Governor of International Council for Computer Communications (ICCC).

His vision is to see all villages in India connected in three years. "We want to see India as a design house and want India to have 200 million telephone and Internet connections. If we are the best, we must deliver." ■



Ashok Jhunjunwala, Head, Department of Electrical Engineering at IIT, Madras, also the driving force behind the innovative use of railway cables to provide Internet connectivity

"We wanted to do something for the country...we see India as a design house"

innovations at TeNeT. The group is also guided by faculty from the departments of Electrical Engineering and Computer Science & Engineering and supported by several researchers, engineers and technical staff

After a B.Tech from IIT, Kanpur in 1975, Jhunjunwala went on to obtain a PhD from the University of Maine. He also served as Assistant Professor at Washington State University. But then India beckoned and since 1981, Jhunjunwala has been a professor at IIT, Madras.

Together with his colleagues, Bhaskar Ramamurti and Timothy Gonsalves, also ex-IIT-ians, he set up TeNeT at IIT, Madras more than 10 years back. Jhunjunwala reminisces, "My colleague Bhaskar was into wireless systems, I had some knowledge of telecom systems and Gonsalves was a software expert. We had an urge to do something for the country. So we started playing